

S181M116 Derivatives and Alternative Investments

Assignment 1: Solution Key

Based on Lecture 1 and Lecture 2, Sections 1–3

Total: 20 points. This solution key corresponds to the Moodle assignment on derivative basics, futures markets, hedging, basis risk, and forward/futures pricing. It excludes the later Lecture 2 material on commodity behavior, stock-index futures and beta hedging, and stack-and-roll hedging.

Question 1: Concepts, Market Structure, and Risk Controls [3 points]

Solution.

- (a) This is a derivative. The underlying variable is the terminal asset price S_T .
 - (b) This is not a derivative. It is the underlying security itself.
 - (c) This is a derivative. The underlying variable is the EUR/USD exchange rate.
 - (d) This is a derivative. The underlying variable is the July temperature index or average temperature measure specified in the contract.
- The natural starting point is an OTC forward contract. The benefit is customization: the contract can specify the exact metal grade, quantity, delivery date, location, and settlement terms. The main risk is that OTC contracts create more bilateral counterparty, liquidity, legal, and valuation risk than standardized exchange-traded futures.
- Separation of front, middle, and back office functions reduces the chance that the same people who take risk also measure, approve, settle, and report that risk. It creates independent checks on trade entry, valuation, limits, collateral, and cash movements.

Question 2: Payoff, Profit, and Option Logic [3 points]

Solution. For the call,

$$\text{call payoff} = \max(S_T - 100, 0), \quad \text{call profit} = \text{call payoff} - 7.$$

For the put,

$$\text{put payoff} = \max(100 - S_T, 0), \quad \text{put profit} = \text{put payoff} - 5.$$

S_T	Call payoff	Call profit	Put payoff	Put profit
88	0	-7	12	7
100	0	-7	0	-5
116	16	9	0	-5

An option can have a positive payoff and still produce a negative profit when the payoff is smaller than the premium paid. For example, a call payoff of 4 would still imply a profit of $4 - 7 = -3$ for this call buyer.

The buyer of a call has limited downside: the most the buyer can lose is the premium. The buyer of a forward has no such premium cap on losses; a long forward loses $K - S_T$ when the terminal spot price is below the delivery price, and that loss can be large.

Question 3: Futures Settlement, Margin, and Liquidity Risk [3 points]

Solution. The trader is short

$$6 \times 5,000 = 30,000$$

bushels. The settlement price increased from 580.25 cents to 592.75 cents, a change of

$$592.75 - 580.25 = 12.50$$

cents per bushel, or USD 0.125 per bushel. A short futures position loses when the futures price rises, so today's futures profit/loss is

$$-30,000 \times 0.125 = -\$3,750.$$

The margin account balance after daily settlement is

$$21,000 - 3,750 = \$17,250.$$

Since USD 17,250 is above the maintenance margin of USD 16,000, there is no margin call. The required deposit is therefore zero.

Daily settlement can create liquidity risk because futures losses must be paid as cash immediately, even if the hedge reduces the firm's total economic risk. The offsetting gain on the underlying exposure may arrive only later, when the physical purchase or sale occurs.

Question 4: Hedge Direction, Basis Risk, and Effective Prices [4 points]

Solution. The firm expects to sell the commodity later. It is hurt by falling spot prices, so it should use a short hedge: losses on the physical sale price are partly offset by gains on the short futures position when futures prices fall.

The gain on the short futures position is

$$F_1 - F_2 = 47.80 - 46.20 = 1.60.$$

The effective sale price is the spot sale price plus the futures gain:

$$S_2 + (F_1 - F_2) = 45.90 + 1.60 = 47.50.$$

The closing basis is

$$b_2 = S_2 - F_2 = 45.90 - 46.20 = -0.30.$$

Therefore

$$F_1 + b_2 = 47.80 - 0.30 = 47.50,$$

which equals the effective sale price.

If the firm originally expected $b_2 = -0.60$, the expected effective sale price was

$$47.80 - 0.60 = 47.20.$$

The realized effective sale price of 47.50 was better by 0.30 per unit. The basis was stronger than expected.

Question 5: Cross Hedging and the Minimum-Variance Hedge Ratio [4 points]

Solution. The minimum-variance hedge ratio is

$$h^* = \rho \frac{\sigma_S}{\sigma_F} = 0.82 \frac{0.24}{0.20} = 0.82(1.20) = 0.9840.$$

The unrounded number of contracts is

$$N^* = h^* \frac{Q_A}{Q_F} = 0.9840 \frac{1,680,000}{42,000} = 0.9840(40) = 39.36.$$

The nearest integer hedge is 39 contracts. A practical recommendation is to use 39 contracts if the objective is to stay closest to the minimum-variance quantity. If the airline is especially concerned about under-hedging fuel price increases, 40 contracts is also defensible, but it is a slight over-hedge relative to the estimated minimum-variance position.

The airline should go long futures because it will buy jet fuel later and is hurt by rising fuel prices. A long futures position gains when the futures price rises.

Residual risk remains because jet fuel and heating oil are different assets, so their prices may not move one-for-one. Other residual risks include timing mismatch, location mismatch, changing correlations, basis risk when the hedge is closed, and contract-size rounding.

Question 6: Forward Pricing and Contract Value [3 points]

Solution. For a stock with known cash income, the no-arbitrage forward price is

$$F_0 = (S_0 - I)e^{rT}.$$

Here $S_0 = 96$, $I = 1.80$, $r = 0.04$, and $T = 0.75$, so

$$F_0 = (96 - 1.80)e^{0.04 \times 0.75} = 94.20e^{0.03} \approx 97.07.$$

The quoted forward price of 99.50 is too high relative to the no-arbitrage price. The cash-and-carry arbitrage is:

1. Buy the stock today.
2. Finance the purchase, taking account of the present value of the known dividends.
3. Short the forward at 99.50.
4. At maturity, deliver the stock into the forward and receive 99.50.

The financing cost net of the known dividends is

$$(S_0 - I)e^{rT} = 97.07.$$

The locked-in profit per share is therefore approximately

$$99.50 - 97.07 = 2.43.$$

The value of an old long forward contract is

$$f = (F_0 - K)e^{-rT}.$$

With $F_0 = 94$, $K = 92$, $r = 0.04$, and $T = 0.5$,

$$f = (94 - 92)e^{-0.04 \times 0.5} = 2e^{-0.02} \approx 1.96.$$

The old long forward is valuable because it allows the holder to buy at 92 when a new forward with the same maturity would lock in a delivery price of 94.